



Some Results of the LEO-II Project¹

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1 Higher-Order Logic (HOL)

The Good Thing: Expressivity

The Bad Thing: Automation is a Challenge

2 The LEO-II Prover

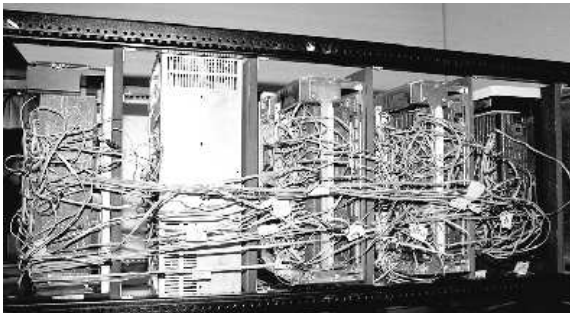
Motivation and Architecture

Solving Lightweight Problems

Solving Less Lightweight Problems

Ongoing and Future Work

Higher-Order Logic (HOL)



Higher-Order Logic (HOL)

Property	FOL	HOL	Example
Quantification over			
- individuals	✓	✓	$\forall x. P(F(x))$
- functions	—	✓	$\forall F. P(F(x))$
- predicates/sets/relations	—	✓	$\forall P. P(F(x))$
Unnamed			
- functions	—	✓	$(\lambda x. x)$
- predicates/sets/relations	—	✓	$(\lambda x. x \neq 2)$
Statements about			
- functions	—	✓	<i>continuous</i> $(\lambda x. x)$
- predicates/sets/relations	—	✓	<i>reflexive</i> $(=)$

HOL is Expressive

$$A \cup B := \{x \mid x \in A \vee x \in B\}$$

$$A \cup B := (\lambda x. x \in A \vee x \in B)$$

$$\cup := \lambda A. \lambda B. (\lambda x. x \in A \vee x \in B)$$

$$\textit{symmetric} := \lambda F. (\forall x, y. F(x, y) = F(y, x))$$

$$\textit{Theorem} : \quad \textit{symmetric}(\cup)$$

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Sets and Relations in HOL

$\in$	$:=$	$\lambda x. \lambda A. A(x)$
$\emptyset$	$:=$	$\lambda x. \perp$
$\cap$	$:=$	$\lambda A. \lambda B. (\lambda x. x \in A \wedge x \in B)$
$\cup$	$:=$	$\lambda A. \lambda B. (\lambda x. x \in A \vee x \in B)$
$\backslash$	$:=$	$\lambda A. \lambda B. (\lambda x. x \in A \vee x \notin B)$
...		
$\subseteq$	$:=$	$\lambda A. \lambda B. (\forall x. x \in A \Rightarrow x \in B)$
$\mathcal{P}$	$:=$	$\lambda A. (\lambda B. B \subseteq A)$
...		
<i>reflexive</i>	$:=$	$\lambda R. (\forall x. R(x, x))$
...		

Without Types: HOL is too Expressive

Russel's Paradox

The set of all sets which do not contain themselves:

$$\{x \mid x \notin x\} \text{ resp. } (\lambda x. x \notin x) \text{ resp. } (\lambda x. \neg x(x))$$

Typed Sets and Relations in HOL

$$\in := \lambda x_{\alpha}. \lambda A_{\alpha \rightarrow o}. A(x)$$

$$\emptyset := \lambda x_{\alpha}. \perp$$

$$\cap := \lambda A_{\alpha \rightarrow o}. \lambda B_{\alpha \rightarrow o}. (\lambda x_{\alpha}. x \in A \wedge x \in B)$$

$$\cup := \lambda A_{\alpha \rightarrow o}. \lambda B_{\alpha \rightarrow o}. (\lambda x_{\alpha}. x \in A \vee x \in B)$$

$$\setminus := \lambda A_{\alpha \rightarrow o}. \lambda B_{\alpha \rightarrow o}. (\lambda x_{\alpha}. x \in A \vee x \notin B)$$

...

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...

Automation of HOL: A Nightmare?

Undecidable and Infinitary Unification

$$\exists F_{\iota \rightarrow \iota}. F(g(x)) = g(F(x))$$

(1) $F \leftarrow \lambda y_i. y$

(2) $F \leftarrow \lambda y_i. g(y)$

(3) $F \leftarrow \lambda y_i. g(g(y))$

(4) ...



Automation of HOL: A Nightmare?

Primitive Substitution

Example Theorem: $\exists S. \text{reflexive}(S)$

Negation and Expansion of Definitions:

$$\neg \exists S. (\forall x. S(x, x))$$

Clause Normalisation ($a(S)$ Skolem term):

$$\neg S(a(S), a(S))$$

Guess some suitable instances for S

$$S \leftarrow \lambda y. \lambda z. \text{True}$$

$$\rightsquigarrow \neg \text{True}$$

$$S \leftarrow \lambda y. \lambda z. V(y, z) = W(y, z)$$

$$\rightsquigarrow V(a(S), a(S)) \neq W(a(S), a(S))$$

$$S \leftarrow \dots$$



Automation of HOL: A Nightmare?

Cut rule

$$\frac{A \Rightarrow C \quad C \Rightarrow B}{A \Rightarrow B}$$

considered as bad in ATP

Calculi that avoid these axioms

- ▶ Axiom of excluded middle ✓
- ▶ Comprehension axioms ✓
- ▶ Functional and Boolean extensionality ✓ [CADE-98, PhD-99]
- ▶ Leibniz and other definitions of equality ✓ [CADE-99, PhD-99]
- ▶ Axiom of induction —
- ▶ Axiom of choice —
- ▶ Axiom of description —

Automation of HOL: A Nightmare?

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[IJCAR-06]: Axioms that imply Cut Calculi that avoid these axioms

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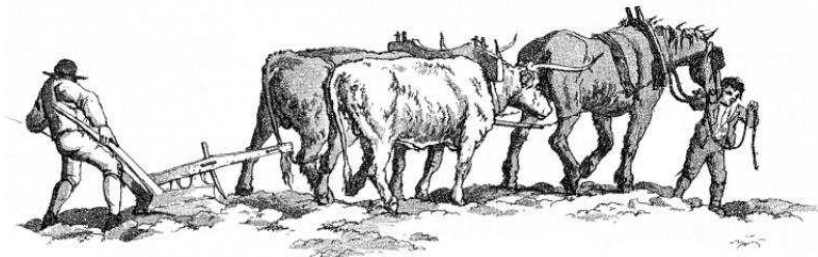
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LEO-II

UNIVERSITY OF CAMBRIDGE
UNIVERSITÄT DES SAARLANDES

An Effective Higher-Order Theorem Prover



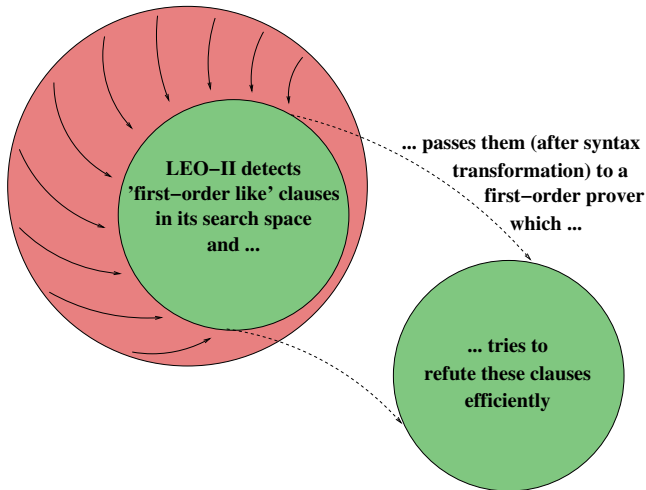
LEO-II employs FO-ATPs:

E, Spass, Vampire

Motivation for LEO-II

- ▶ TPS system of Peter Andrews et al.
- ▶ LEO hardwired to Ω_{MEGA} (predecessor of LEO-II)
[CADE-98, PhD-99]
- ▶ Agent-based architecture $\Omega\text{-ANTS}$
(with V. Sorge) [AIMSA-98, EPIA-99, Calculemus-00]
- ▶ Collaboration of LEO with FO-ATP via $\Omega\text{-ANTS}$
(with V. Sorge) [KI-01, LPAR-05, JAL-07]
- ▶ Progress in Higher-Order Termindexing
(with F. Theiss and A. Fietzke) [IWIL-06]

Architecture of LEO-II



Solving Lightweight Problems



Example: TPTP Problem SET171+3

Axiomatization in FO Set Theory

Assumptions:

$$\forall B, C, x. (x \in (B \cup C) \Leftrightarrow x \in B \vee x \in C)$$

$$\forall B, C, x. (x \in (B \cap C) \Leftrightarrow x \in B \wedge x \in C)$$

$$\forall B, C. (B \subseteq C \Leftrightarrow \forall x. x \in B \Rightarrow x \in C)$$

$$\forall B, C. (B \cup C = C \cup B)$$

$$\forall B, C. (B \cap C = C \cap B)$$

$$\forall B, C. (B = C \Leftrightarrow B \subseteq C \wedge C \subseteq B)$$

$$\forall B, C. (B = C \Leftrightarrow \forall x. x \in B \Leftrightarrow x \in C)$$

Proof Goal:

$$\forall B, C, D.$$

$$B \cup (C \cap D) = (B \cup C) \cap (B \cup D)$$

Performance: FO-ATPs

```
% SPASS---3.0
% Problem : SET171+3
% SPASS beiseite: Ran out of time.
% E---0.999
% Problem : SET171+3
% Failure: Resource limit exceeded
(time)
% Vampire---9.0
% Problem : SET171+3
% Result : Theorem 68.6s
```

Performance: LEO-II + E

```
Eureka --- Thanks to Corina!
Total Reasoning Time: 0.03s
LEO-II (Proof Found!)
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Results

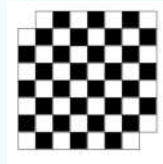
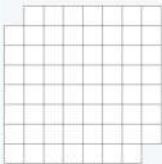
Problem	Vamp. 9.0	LEO+Vamp.	LEO-II+E
014+4	114.5	2.60	0.300
017+1	1.0	5.05	0.059
066+1	—	3.73	0.029
067+1	4.6	0.10	0.040
076+1	51.3	0.97	0.031
086+1	0.1	0.01	0.028
096+1	5.9	7.29	0.033
143+3	0.1	0.31	0.034
171+3	68.6	0.38	0.030
580+3	0.0	0.23	0.078
601+3	1.6	1.18	0.089
606+3	0.1	0.27	0.033
607+3	1.2	0.26	0.036
609+3	145.2	0.49	0.039
611+3	0.3	4.00	0.125
612+3	111.9	0.46	0.030
614+3	3.7	0.41	0.060
615+3	103.9	0.47	0.035
623+3	—	2.27	0.282
624+3	3.8	3.29	0.047
630+3	0.1	0.05	0.025
640+3	1.1	0.01	0.033
646+3	84.4	0.01	0.032
647+3	98.2	0.12	0.037

Problem	Vamp. 9.0	LEO+Vamp.	LEO-II+E
648+3	98.2	0.12	0.037
649+3	117.5	0.25	0.037
651+3	117.5	0.09	0.029
657+3	146.6	0.01	0.028
669+3	83.1	0.20	0.041
670+3	—	0.14	0.067
671+3	214.9	0.47	0.038
672+3	—	0.23	0.034
673+3	217.1	0.47	0.042
680+3	146.3	2.38	0.035
683+3	0.3	0.27	0.053
684+3	—	3.39	0.039
716+4	—	0.40	0.033
724+4	—	1.91	0.032
741+4	—	3.70	0.042
747+4	—	1.18	0.028
752+4	—	516.00	0.086
753+4	—	1.64	0.037
764+4	0.1	0.01	0.032

Vamp. 9.0: 2.80GHz, 1GB memory, 600s time limit
LEO+Vamp.: 2.40GHz, 4GB memory, 120s time limit
LEO-II+E: 1.60GHz, 1GB memory, 60s time limit

Representation (and the right System Architecture) Matters!

A general lesson in AI ...

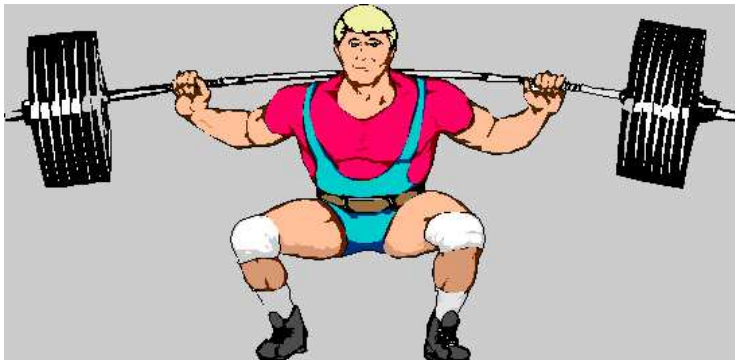


...and a specific lesson here

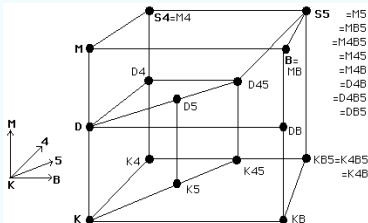
FOL
+
FO-ATP

HOL
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LEO-II + FO-ATP

Solving Less Lightweight Problems



Modal Logics Challenge



John Halleck (U Utah):
<http://www.cc.utah.edu/~nahaj/>
 \$100 Modal Logic Challenge:
www.tptp.org

Example

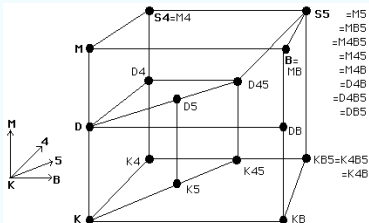
$$\begin{aligned}
 S4 &= K \\
 &+ M: \Box_R A \Rightarrow A \\
 &+ 4: \Box_R A \Rightarrow \Box_R \Box_R A
 \end{aligned}$$

Theorems:

$$\begin{aligned}
 S4 &\not\subseteq K & (1) \\
 (M \wedge 4) &\Leftrightarrow (refl(R) \wedge trans(R)) & (2)
 \end{aligned}$$

Experiments

	FO-ATPs [SutcliffeEtal-07]	LEO-II + E [BePa-08]
(1)	16min + 2710s	17.3s
(2)	???	2.3s



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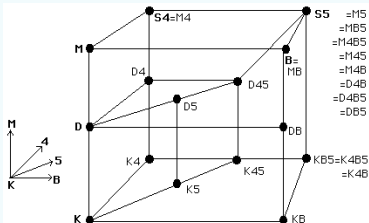
$$\begin{array}{lcl} S4 & = & K \\ + & M: & \Box_R A \Rightarrow A \\ + & 4: & \Box_R A \Rightarrow \Box_R \Box_R A \end{array}$$

Theorems:

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Much simpler: Reasoning within Multimodal Logics

Problem	LEO-II + E
$\text{valid}(\Box_r \top)$	0.025s
$\text{valid}(\Box_r a \Rightarrow \Box_r a)$	0.026s
$\text{valid}(\Box_r a \Rightarrow \Box_s a)$	—
$\text{valid}(\Box_s (\Box_r a \Rightarrow \Box_r a))$	0.026s
$\text{valid}(\Box_r (a \wedge b) \Leftrightarrow (\Box_r a \wedge \Box_r b))$	0.044s
$\text{valid}(\Diamond_r (a \Rightarrow b) \Rightarrow \Box_r a \Rightarrow \Diamond_r b)$	0.030s
$\text{valid}(\neg \Diamond_r a \Rightarrow \Box_r (a \Rightarrow b))$	0.029s
$\text{valid}(\Box_r b \Rightarrow \Box_r (a \Rightarrow b))$	0.026s
$\text{valid}((\Diamond_r a \Rightarrow \Box_r b) \Rightarrow \Box_r (a \Rightarrow b))$	0.027s
$\text{valid}((\Diamond_r a \Rightarrow \Box_r b) \Rightarrow (\Box_r a \Rightarrow \Box_r b))$	0.029s
$\text{valid}((\Diamond_r a \Rightarrow \Box_r b) \Rightarrow (\Diamond_r a \Rightarrow \Diamond_r b))$	0.030s

(Normal) Multimodal Logic in HOL

Simple, Straightforward Encoding of Multimodal Logic

- ▶ base type ι : set of possible worlds
- certain terms of type $\iota \rightarrow o$: multimodal logic formulas
- ▶ multimodal logic operators:

$$\begin{aligned}
 \neg (\iota \rightarrow o) \rightarrow (\iota \rightarrow o) &= \lambda A_{\iota \rightarrow o}. (\lambda x_{\iota}. \neg A(x)) \\
 \vee (\iota \rightarrow o) \rightarrow (\iota \rightarrow o) \rightarrow (\iota \rightarrow o) &= \lambda A_{\iota \rightarrow o}. \lambda B_{\iota \rightarrow o}. (\lambda x_{\iota}. A(x) \vee B(x)) \\
 \Box_R (\iota \rightarrow \iota \rightarrow o) \rightarrow (\iota \rightarrow o) \rightarrow (\iota \rightarrow o) &= \lambda R_{\iota \rightarrow \iota \rightarrow o}. \lambda A_{\iota \rightarrow o}. \\
 &\quad (\lambda x_{\iota}. \forall y_{\iota}. R(x, y) \Rightarrow A(y))
 \end{aligned}$$

Related Work

[Gallin-73], [Carpenter-98], [Merz-99],
[\[Brown-05\]](#), [Hardt&Smolka-07], [Kaminski&Smolka-07]

(Normal) Multimodal Logic in HOL

Encoding of Validity

$$\text{valid} \quad := \quad \lambda A_{\iota \rightarrow o}. (\forall w_{\iota}. A(w))$$

Encoding of the Theorems

$$(1) \quad \forall R. (\forall A. \text{valid}(\Box_R A \Rightarrow A) \wedge \text{valid}(\Box_R A \Rightarrow \Box_R \Box_R A)) \\ \Leftrightarrow (\text{reflexive}(R) \wedge \text{transitive}(R))$$

(2.3s, LEO-II passes 70 clauses to E, E generates 21769 clauses before finding the empty clause)

$$(2) \quad \exists R. \exists A. \exists B. (\neg \text{valid}(\Box_R A \Rightarrow A)) \vee (\neg \text{valid}(\Box_R B \Rightarrow \Box_R \Box_R B))$$

(17.3s, LEO-II instantiates R with $\neq$ via primitive substitution)

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Limitations

LEO-II cannot prove the following example:

Modal logic $K4$ (which adds only axiom 4 to K) is not entailed in K :

$$\exists R. \exists B. (\neg \text{valid}(\Box_R B \Rightarrow \Box_R \Box_R B))$$

LEO-II also cannot prove this related example:

$$\exists R. \neg \text{trans}(R)$$

Reason: not a theorem; domain of possible worlds may well just consist of a single world w .

LEO-II can in fact prove the latter example under the additional assumption

$$\exists X. \exists Y. X \neq Y$$

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valid($\Box_s (\Box_r a \Rightarrow \Box_r a)$)

Initialisation of problem

$$\neg \text{valid}(\Box_s (\Box_r a \Rightarrow \Box_r a))$$

Definition expansion

$$\neg (\forall x_{\iota}, \forall y_{\iota}, : \neg s(x, y) \vee ((\neg (\forall u_{\iota}, \neg r(y, u) \vee a(u))) \vee (\forall v_{\iota}, \neg r(y, v) \vee a(v)))$$

Normalisation (x, y, u are now Skolem constants, V is a variable)

$$\begin{array}{ll} s(x, y) & \neg a(u) \\ r(y, u) & a(V) \vee \neg r(y, V) \end{array}$$

Translation to first-order logic

$$\begin{array}{ll} @_{(io)_i}(@_{(i(io))_i}(s, x), y) & \neg @_{(io)_i}(a, u) \\ @_{(io)_i}(@_{(i(io))_i}(r, y), u) & @_{(io)_i}(a, V) \vee \neg @_{(io)_i}(@_{(i(io))_i}(r, y), V) \end{array}$$

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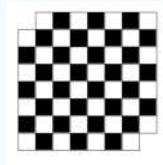
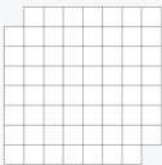
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Representation (and the right System Architecture) Matters!

A general lesson in AI ...



...and a specific lesson here

FOL
+
FO-ATP

HOL
+
LEO-II + FO-ATP

... there is much left to be done!

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- ▶ Equational Reasoning
- ▶ Termination
- ▶ Handling of Definitions

Cooperat. with Specialist Reasoners

- ▶ Monadic Second-Order Logic, Prop. Logic, Arithmetic, ...
- ▶ Logic Translations
- ▶ Feedback for LEO-II
- ▶ Proof Transf./Verification
- ▶ Agent-based Architecture

Integration into Proof Assistants

- ▶ Relevance of Axioms
- ▶ Proof Transf./Verification

International Infrastructure

- ▶ TPTP Language(s) for HOL
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Applications

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Ontology Reasoning (SUMO, CYC),
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Dream or Nightmare?

